

Human-Centered Evaluation and Auditing of Language Models

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Abstract

The recent advancements in Large Language Models (LLMs) have significantly impacted numerous, and will impact more, real-world applications. However, these models also pose significant risks to individuals and society. To mitigate these issues and guide future model development, responsible evaluation and auditing of LLMs are essential. This workshop aims to address the current “evaluation crisis” in LLM research and practice by bringing together HCI and AI researchers and practitioners to rethink LLM evaluation and auditing from a human-centered perspective. The workshop will explore topics around understanding stakeholders’ needs and goals with evaluation and auditing LLMs, establishing human-centered evaluation and auditing methods, developing tools and resources to support these methods, building community and fostering collaboration. By soliciting papers, organizing invited keynote and panel, and facilitating group discussions, this workshop aims to develop a future research agenda for addressing the challenges in LLM evaluation and auditing. Following a successful first iteration of this workshop at CHI 2024, we introduce the theme of “mind the context” for this second iteration, where participants will be encouraged to tackle the challenges and nuances of LLM evaluation and auditing in specific contexts.

CCS Concepts

• Human-centered computing → HCI design and evaluation methods; HCI theory, concepts and models; • Computing methodologies → Artificial intelligence.

Keywords

Evaluation, Audit, Large Language Models, Generative AI

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1 Introduction

The recent development of Large language models (LLMs), such as GPT-X models, exhibit impressive and versatile capabilities. They are poised to significantly impact the technology landscape, powering numerous applications in various domains. Meanwhile, these models also raise significant risks for individuals and society [51]. They are known to “hallucinate” information—producing plausible but factually incorrect information and potentially contributing to the spread of misinformation. They can generate harmful, sometimes toxic content. They are found to encode biases, which can propagate harmful discrimination, stereotypes, and exclusion [15]. They may also produce inappropriate content or work ineffectively in some contexts or for some user groups more than others.

The key to mitigating the risks of AI technologies is to perform responsible evaluation and auditing. Evaluation and auditing can be performed by different stakeholders [31]: LLM developers must assess whether the model is ready for deployment, and understand

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when and how the model may fail to enact safe usage policies. Application developers and product managers need the model evaluation results to responsibly decide whether and how to use an LLM. They may also need to perform their own evaluation as they adapt the LLM (i.e., through prompting) or integrate the LLM into their application. End-users and impacted stakeholder groups should be able to assess the model to adapt interactions, provide feedback, and contest the model. Policy-makers and third-party auditors must have means to gauge the model's limits and risks to have oversight.

However, LLMs are giving rise to an “evaluation crisis” in both AI research and practice. While the AI and NLP communities have traditionally relied on benchmarks to perform automated evaluation for specialized models (e.g., machine translation, summarization, QA models), these benchmarks suffer from many limitations in terms of validity and quality to begin with [18], and they are not equipped to handle the diverse, complex, and uncertain capabilities of LLMs. While initial research efforts in creating LLM-specific meta-benchmarks that can capture multiple model capabilities have emerged [19, 28], their ability to predict LLMs’ utility for different downstream applications and stakeholder groups remains unclear. Researchers have also grappled with other fundamental challenges to the evaluation of LLMs, such as output variability and sensitivity to prompts, and unknown data contamination (i.e., the model could have seen a given evaluation test in its training data).

In practice, some forms of human evaluation such as rating outputs by domain experts or crowd workers are often used to evaluate language models. However, these practices are largely ad-hoc and have been widely criticized for lacking standardization, reproducibility, and validity for assessing model utility in real-world settings [6]. It is also known that practitioners engage in extensive internal auditing (e.g., red teaming [16]) to mitigate the safety concerns and other risks of LLMs, although there is a lack of transparency and shared practices in how such audits are performed [39, 43, 44, 46]. Moreover, the fundamentally socio-technical and contested nature of AI fairness and transparency adds unique challenges in conducting human-centered evaluation and auditing of LLMs [17, 40, 47, 57]. For example, HCI researchers have surfaced challenges around the potential blindspots when the evaluation and auditing of AI systems are only conducted by small groups of technical experts, without meaningful engagements from diverse stakeholders [11, 22, 36, 56].

To incorporate the perspectives and lived experiences of diverse stakeholders in AI evaluation and auditing, there’s growing interest among both HCI and AI researchers in directly involving these stakeholders in the evaluation and auditing of LLMs (as well as other AI systems) [3, 13, 23, 26, 41, 42, 45]. In parallel with these research efforts, many major technology companies have begun to experiment with approaches that directly engage users in evaluating and auditing their LLM-powered products and services for problematic behaviors. For instance, OpenAI, inspired by Twitter’s “algorithmic bias bounty,” [5], initiated a series of feedback contests and red teaming efforts to engage end users in providing feedback on problematic model outputs during their interactions with the ChatGPT [1]. However, AI practitioners often face practical challenges when conducting LLM audits, such as identifying and

recruiting the right groups of stakeholders [7, 12], effectively guiding users in auditing LLMs [7, 45], and aggregating the auditing reports from a vast user base [3, 7].

Given these challenges, there is a growing call for rethinking *what* to evaluate and *how* to evaluate & audit LLMs. This workshop aims to bring together HCI and AI researchers and practitioners to contemplate these questions by taking human-centered perspectives, starting by asking: *Who* should be involved in evaluating and auditing LLMs? What are their needs and goals? What are their constraints and challenges? How would they make use of the evaluation and auditing results? Furthermore, we believe it is timely to bring HCI lessons and perspectives about technology evaluation—a core focus of the HCI discipline—into shaping LLM evaluation practices [32]. These perspectives may include a shift away from automatic evaluation to human-subjects evaluation; embracing diverse quantitative and qualitative, empirical and analytical, evaluation methods; considerations of what constitutes “good” or appropriate evaluation methods; best practices for designing evaluation studies and auditing pipelines; tools, processes, and resources (e.g., datasets, statistical tools) to support LLM evaluation practices; regulations and policies that could prioritize marginalized community members by giving them a greater voice in the LLM audits, and many more.

In summary, this workshop aims to solicit position papers, organize an invited panel, and bring together workshop attendees for group discussions to share research and knowledge about, and develop a future research agenda for, the following topics:

- Identifying current challenges in LLM evaluation and auditing and reasons behind them.
- Understanding the role of various stakeholders in LLM evaluation and auditing.
- Establishing human-centered evaluation and auditing methods of LLMs.
- Developing tools, processes, and resources to support stakeholders to evaluate and audit LLMs.
- Discussing regulatory measures, public policies, or industry standards around evaluating and auditing LLMs.
- Sharing perspectives about the *what*, *why*, and *how* of LLM evaluation and auditing, and opportunities for HCI research and collaboration between the HCI and AI communities.

Theme: Mind the Context

The first iteration of this workshop at CHI 2024 [52]¹ received 38 paper submissions and welcomed 45 attendees from 30 different institutions around the globe. Following this success, we propose the theme of “mind the context” for this second iteration to encourage attendees to engage with specific contexts in LLM evaluation and auditing. This theme involves various topics: the usage contexts of LLMs (e.g., evaluating the capabilities and limitations of LLM applications in mental wellness care, or translation in high-stakes scenarios), the context of the evaluation/auditing itself (e.g., who are using LLM evaluation tools, and how should we design these tools with this context in mind?), and more. We purposefully leave “context” open for interpretation by participants, so to encourage

¹<https://heal-workshop.github.io/>

diversity in how participants conceptualize and operationalize this key concept in LLM evaluation and auditing.

2 Organizing Committee

This workshop is organized by a team of experienced researchers with diverse research backgrounds in model evaluation and auditing and at different stages of their research careers. Collectively, members of the organizing committee contributed to HCI and AI on various topics, including Explainable AI [29, 31, 58], Responsible AI [7, 9, 10, 30, 35, 48–50], Language Technologies [2, 8, 13, 24, 38, 54, 59], Model Evaluation [4, 20, 21, 25, 32–34, 37, 53, 55], and Algorithm Auditing [7, 11, 14, 26, 27]. Members of the organizing committee have successfully hosted various workshops, tutorials, and panels on the topics of human-centered AI, Algorithm Testing and Auditing, and Evaluation of NLP systems, starting important conversations and community buildings around human-centered AI evaluation and auditing. By integrating expertise from both HCI and AI, as well as our extensive prior experiences around organizing workshops, we aim to bridge the gap between HCI and AI, promote effective discussion, facilitate interdisciplinary collaboration, and galvanize community building.

Yu Lu Liu is a Ph.D. student at Johns Hopkins University. She is interested in NLP evaluation and responsible AI. Her past work examined the design of NLP benchmarks and responsible AI practices in automatic text summarization.

Wesley Hanwen Deng is a Ph.D. student at the Human-Computer Interaction Institute (HCII) at Carnegie Mellon University. Wesley's current works aim to support responsible AI practice in real-world contexts, including scaffolding effective communication and collaborations between AI developers and users. Before joining CMU, Wesley completed his BA in Computer Science with the Highest Honor at UC Berkeley, where he also received an EECS research honor. Wesley led the efforts to organize a Workshop on "User Engagement in Algorithm Testing and Auditing" at FAccT 2023, a Workshop on "Supporting User Engagement in Testing, Auditing, and Contesting AI" at CSCW 2023,² a Workshop on "Responsible Crowdsourcing for Responsible Generative AI" at HCOMP 2024, a Special Interest Group on "Collaboratively Designing and Evaluating Responsible AI Interventions" at CSCW 2024, as well as the first iteration of this workshop at CHI 2024.

Michelle S. Lam is a Ph.D. candidate in Computer Science at Stanford University in the Human-Computer Interaction Group. She builds systems for non-AI experts to take creative control over AI by allowing them to define AI behavior in their own words—from objective functions to evaluations. Her work on novel methods for human-centered machine learning and algorithm auditing has received awards at top human-computer interaction venues like CHI and CSCW. She is supported by a Stanford Interdisciplinary Graduate Fellowship and has been a Stanford HAI Graduate Fellow, Stanford Technology and Racial Equity Graduate Fellow, Brown Institute for Media Innovation Magic Grantee, and Siebel Scholar. Michelle co-organized the first iteration of this workshop at CHI 2024 and co-organized a CSCW 2023 workshop on user engagement in AI auditing.

Motahhare Eslami is an assistant professor at the School of Computer Science, Human-Computer Interaction Institute, and Software and Societal Systems Department, at Carnegie Mellon University. Motahhare's research goal is to investigate the existing accountability challenges such as auditing in algorithmic systems and to empower the users of algorithmic systems, particularly those who belong to marginalized communities and those whose decisions impact marginalized communities, make transparent, fair, and informed decisions in interaction with algorithmic systems. Motahhare is named one of the 100 Brilliant Women in AI Ethics. Her work has been recognized with a Google Ph.D. Fellowship, Best Paper and Honorable Mention Awards, as well as award for contribution to inclusion and diversity at top-tier ACM conferences. She co-organized the first iteration of this workshop at CHI 2024 as well as a CSCW 2023 workshop on user engagement in AI auditing.

Juho Kim is an Associate Professor in the School of Computing at KAIST, affiliate faculty in the Kim Jaechul Graduate School of AI at KAIST, and a director of KIXLAB (the KAIST Interaction Lab). His research in human-computer interaction and human-AI interaction focuses on building interactive and intelligent systems that improve the ways people learn, work, create, and collaborate. He gave a keynote at NeurIPS 2022 titled "Interaction-Centric AI". He co-organized the first iteration of this workshop at CHI 2024.

Q. Vera Liao is a Principal Researcher at Microsoft Research Montreal, where she is part of the FATE (Fairness, Accountability, Transparency, and Ethics) group. Her current research interests are in human-AI interaction, explainable AI, and responsible AI, with an overarching goal of bridging emerging AI technologies and human-centered research and design practices. Prior to joining MSR, she worked at IBM Research, and studied at the University of Illinois at Urbana-Champaign and Tsinghua University. She has co-organized many workshops at CHI, IUI and NeurIPS, including CHI 2021 and 2022 Workshops on Human-Centered Explainable AI, CHI 2023 Workshop on GenAI and HCI, as well as the first iteration of this workshop at CHI 2024.

Wei Xu is an Associate Professor in the College of Computing at Georgia Tech. Her recent research focuses on analyzing and evaluating large language models. She is a recipient of the NSF CAREER Award, best paper awards and honorable mentions from COLING '18, ACL '23, and ACL '24. She is an NAACL executive board member and regularly serves as a (senior) area chair for *ACL conferences. She has co-founded and co-organized multiple workshops, including WNUT, GEM, and TSAR.

Jekaterina Novikova is a Science Lead at the AI Risk and Vulnerability Alliance. Her research focuses on evaluating large language models for interdisciplinary applications, including healthcare and human-robot interaction. She was co-organizing multiple workshops, including GEMv2 at EMNLP'22 and ML4CMH at AAAI'24.

Ziang Xiao is an assistant professor in the Department of Computer Science. His work lies in the intersection of human-computer interaction, natural language processing, and social and personality psychology. Ziang is on the organizing committee and an associate chair for multiple HCI venues (CHI, CSCW, IUI). He co-organized the 3rd HCI+NLP workshop at NAACL 2024. He co-organized the first iteration of this workshop at CHI 2024.

²<https://cscw-user-ai-auditing.github.io/>

3 Plans to Publish Workshop Proceedings

To find suitable attendees (approximately 50-60 participants), we will ask participants to submit a position paper or short paper with preliminary findings (up to four pages) on human-centered evaluation and auditing of LLMs. Hence, the workshop organizers will put out a call to solicit papers and form a program committee to review and carefully select the incoming submissions. We plan to provide at least two reviews for each submission to assess the quality and relevance of participants' content. The authors of selected submissions will be invited to participate in our workshop. This way, we will control the number of participants at our workshop to avoid physical space shortages.

The primary aim of this workshop is to build a new research community on the topic to share future research agendas and promote collaboration among researchers from various backgrounds and viewpoints. All accepted papers from attendees will be published on the workshop's website to guarantee widespread distribution of the opinions and findings. Furthermore, the accepted papers will be featured in the workshop proceedings, which we hope to publish on *ceur-ws*.³ Finally, we will actively encourage all authors to submit extended versions of their papers for potential inclusion in a special issue in a well-recognized HCI journal.

Ideas and discussion topics will be collected throughout the session in a shared document, both during and after the workshop. Attendees will also be invited to participate in future joint projects stemming from the discussion and presentations at the workshop.

4 In-person and Hybrid

We acknowledge the numerous challenges and concerns that attendees may face, including visa complications, financial limitations, and personal obligations, which could potentially prevent them from participating in an in-person workshop. Since the main goal for this workshop is community building, we believe it is crucial to make our workshop open and inclusive for everyone. Therefore, we plan to host the workshop in a hybrid form. We are dedicated to supporting the CHI's initiative to foster an inclusive environment for both in-person and hybrid attendees.⁴ We will facilitate synchronous discussions for both in-person attendees with dedicated cameras and remote attendees on Zoom. We will utilize online platforms like Miro to create a coherent experience for both in-person and remote attendees during the discussion sessions. All attendees will use the same Slack channel to share workshop materials and connect with each other. All keynote, panels, and presentations will be live for remote participants with live captioning.

5 Asynchronous Engagement

We will utilize email lists, social media and workshop websites to encourage participation and share key information. The website will function as a central place for distributing information, such as the workshop description and proposal, a call for participation, the workshop schedule, and information about the keynote speaker and panelists.

We will create a dedicated Slack group to facilitate asynchronous engagement. Before the workshop, all registered attendees

will be invited to the Slack group for pre-workshop activities. We will survey attendees to ask about their interests in this topic and ideal takeaways. During the workshop, the Slack channel will be used to disseminate important information and share workshop materials. After the workshop, the Slack group will be actively maintained by the organizers for attendees to further engage with each other, discuss related issues, share research outcomes, and incubate collaborations.

We will strongly encourage attendees to create a 5-minute video presentation for their accepted workshop paper, which will be shown on the workshop's website with captions. We will also record the keynote and panel talk during the workshop. Attendees can access the video recordings asynchronously on Slack.

6 Workshop Structure

We design a workshop structure for the goal of building community, exchanging knowledge and co-creating new research agendas for this emerging topic. We plan for a one-day, in-person/hybrid workshop structured in a series of presentations, activities, and guided discussions. We are aiming for approximately 50-60 attendees. Given we will use our pre-workshop survey to guide the exact topics for the discussion section, our tentative schedule is below.

- (1) **Welcome (15 mins):** Opening talk by one of the organizers, with brief introductions from organizers, highlighting the workshop's goals and going over plans for the day.
- (2) **Opening Keynote (45 mins):** A distinguished keynote speaker will deliver a 30-minute opening keynote with their insights and perspectives on human-centered evaluation and auditing of LLMs, focusing on the workshop's theme of "mind the context". Following the presentation, there will be a 15-minute Q&A session, providing an opportunity for attendees to engage with the speaker and ask questions related to the topic.
- (3) **Coffee Break (15 mins)**
- (4) **Author Presentation I (60 mins):** Authors of accepted papers will give a 5-minute lightning talk about their paper. At the end, we will have 15 minutes to invite authors as a panel to answer the audience's questions.
- (5) **Discussion Session I - Challenges Identification (60 mins):** Attendees will split into smaller groups and discuss the main challenges and goals in LLM evaluation and auditing and their impact on stakeholders in downstream applications. Each group will identify real-world challenges for LLM evaluation and auditing and discuss how current evaluation and auditing practices fell short. We will have designated organizers to facilitate an online discussion session for remote attendees.
- (6) **Lunch Break (90 mins)**
- (7) **Panel (60 mins):** Our panel will consist of senior researchers and practitioners working on human-centered evaluation and auditing of LLMs. The panel will kick off the workshop by discussing different aspects of LLM evaluation and auditing, such as unique challenges in specific contexts, novel evaluation and audit methods for LLMs, and bridging HCI and AI communities on shared challenges. This panel will set a solid foundation for the upcoming activities.

³<https://ceur-ws.org/>

⁴<https://chi2025.acm.org/accessibility-faq/>

- (8) **Author Presentation II, Workshop Theme (60 mins):** Authors of accepted papers which are under the workshop's theme of "mind the context" will give a 5-minute lighting talk about their paper. At the end, we will have 15 minutes to invite authors as a panel to answer the audience's questions.
- (9) **Coffee Break and Poster Session (75 mins):** A long coffee break will co-occur with a poster session so attendees could discover others' work in a relaxed atmosphere. The poster session will give special spotlight to papers under the workshop theme of "mind the context".
- (10) **Discussion Session II - Methods and Tooling (60 mins):** In this session, attendees will explore various design techniques and methodologies that can be employed in LLM evaluation and auditing, taking into account specific contexts. Groups will brainstorm and discuss creative approaches to improve the evaluation process and auditing practice. Discussions will focus on identifying appropriate methods for evaluation goals and exploring the trade-offs between various methods. We will have designated organizers to facilitate an online discussion session for remote attendees.
- (11) **End Note (30 mins):** The organizers will end the workshop by synthesizing the main points from two discussions in previous discussion sessions. Attendees and organizers will reflect together to develop tangible future research topics.

The workshop will alternate between lighting talks, Q&A, group activities, and plenary discussions. The current discussion topics are tentative, and we will change them based on the attendees' interests in the pre-workshop survey. The lighting talks order will also be arranged based on the discussion topic. We will facilitate different groups of attendees interaction to provide attendees with ample opportunities to exchange ideas and form new collaborations.

7 Post-Workshop Plans

The expected workshop outcomes include:

- An active growing community of researchers who are interested in working on human-centered evaluation and auditing for LLMs to exchange ideas and form collaborations. We plan to create email list and a Slack channel to further engage with participants post-workshop.
- An article for the ACM Interactions magazine or a blog post to summarize the workshop outcome and future research landscape for both AI and HCI audiences.
- A workshop proceedings with accepted papers published on *ceur-ws*.⁵
- An invitation to authors of selected accepted papers for a special issue at a well-regarded HCI journal (e.g. TiiS, TOCHI).

8 Accessibility

We will survey attendees before the workshop about their accessibility needs and attending constraints. We will provide essential materials and tools such as large-print handouts, assistive listening devices, and accessible seating to ensure an inclusive design workshop experience for all attendees. We will strongly encourage all attendees to adhere to the ACM's Accessible Submission Guide⁶

when preparing their manuscript and encourage them to perform an accessibility check for the final camera-ready version. To guarantee an inclusive experience for in-person and hybrid attendees, we will ensure automatic transcription services are available for panel and lighting talks. Additionally, to promote accessibility and maintain thorough documentation, assigned note-takers will record crucial points in a shared Google Doc, enabling attendees to revisit discussions and input after the workshop. With conference organizers' support, we will incorporate diverse engagement methods, including video, audio, and text, to accommodate attendees with varied visual, hearing, speech, and cognitive abilities.

9 Call for Participation

As Large Language Models (LLMs) continue to advance, addressing the "evaluation crisis" and exploring human-centered evaluation and auditing approaches become increasingly important. Ensuring that these models are effective, ethical, and socially responsible is crucial for their widespread adoption and integration into many real-world applications. We invite authors from all disciplines to submit papers to the HEAL@CHI'25: Human-centered Evaluation and Auditing of Language Models⁷.

This workshop aims to foster discussions on understanding real-world stakeholders' needs and goals, establishing human-centered evaluation and auditing methods for LLMs, and developing tools, processes, and resources to support stakeholders in evaluating and auditing LLMs.

Papers can address topics, but not limited to, such as:

- Empirical understanding of stakeholders' needs and goals of LLM evaluation and auditing
- Human-centered evaluation and auditing methods for LLMs
- Tools, processes, and guidelines for LLM evaluation and auditing
- Discussion of regulatory measures and public policies for LLM auditing
- Ethics in LLM evaluation and auditing

Theme: Mind the Context We invite authors to engage with specific contexts in LLM evaluation and auditing. This theme could involve various topics: the usage contexts of LLMs (e.g., evaluating the capabilities and limitations of LLM applications in mental wellness care, or translation in high-stakes scenarios), the context of the evaluation/auditing itself (e.g., who are using LLM evaluation tools, and how should we design these tools with this context in mind?), and more. The term "context" is left open for interpretation, so to encourage diversity in how this key concept is conceptualized and operationalized by workshop participants. Papers under this theme will be given a dedicated lightning talk session, as well as a special spotlight during the workshop's poster session.

Submission Requirements: Position papers should be up to 4 pages in the CHI Publication Format, excluding references. Submit papers to [Open Review Link].⁸ Submissions will go through a jury review process and be selected based on quality. Accepted papers will be featured on our website and presented during the workshop, and at least one author must attend the workshop.

⁵<https://ceur-ws.org/>

⁶<https://sigchi.org/conferences/author-resources/accessibility-guide/>

⁷<https://heal-workshop.github.io/>

⁸We plan to use OpenReview to manage the paper submission process.

References

- [1] Open AI. 2022. ChatGPT Feedback Contest: Official Rules. https://cdn.openai.com/chatgpt/ChatGPT_Feedback_Contest_Rules.pdf
- [2] Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. 2021. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258* (2021).
- [3] Ángel Alexander Cabrera, Abraham J Druck, Jason I Hong, and Adam Perer. 2021. Discovering and validating ai errors with crowdsourced failure reports. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–22.
- [4] Ángel Alexander Cabrera, Erica Fu, Donald Bertucci, Kenneth Holstein, Ameet Talwalkar, Jason I Hong, and Adam Perer. 2023. Zeno: An interactive framework for behavioral evaluation of machine learning. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [5] Rumman Chowdhury and Jutta Williams. 2021. Introducing Twitter's first algorithmic bias bounty challenge. URL: https://blog.twitter.com/engineering/en_us/topics/insights/2021/algorithmic-bias-bountychallenge (2021).
- [6] Elizabeth Clark, Tal August, Sofia Serrano, Nikita Haduong, Suchin Gururangan, and Noah A Smith. 2021. All That's 'Human' Is Not Gold: Evaluating Human Evaluation of Generated Text. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 7282–7296.
- [7] Wesley Hanwen Deng, Boyuan Guo, Alicia Devrio, Hong Shen, Motahhare Eslami, and Kenneth Holstein. 2023. Understanding Practices, Challenges, and Opportunities for User-Engaged Algorithm Auditing in Industry Practice. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [8] Wesley. H. Deng, Nikita Mehndru, Samantha Robertson, and Niloufar Salehi. 2022. Beyond General Purpose Machine Translation: The Need for Context-specific Empirical Research to Design for Appropriate User Trust. *Workshop on Trust and Reliance in AI-Human Teaming at CHI '22* (2022). <https://arxiv.org/abs/2205.06920>
- [9] Wesley Hanwen Deng, Manish Nagireddy, Michelle Seng Ah Lee, Jatinder Singh, Zhiwei Steven Wu, Kenneth Holstein, and Haiyi Zhu. 2022. Exploring How Machine Learning Practitioners (Try To) Use Fairness Toolkits. In *2022 ACM Conference on Fairness, Accountability, and Transparency*. ACM, Seoul Republic of Korea, 473–484. <https://doi.org/10.1145/3531146.3533113>
- [10] Wesley Hanwen Deng, Nur Yildirim, Monica Chang, Motahhare Eslami, Kenneth Holstein, and Michael Madaio. 2023. Investigating Practices and Opportunities for Cross-functional Collaboration around AI Fairness in Industry Practice. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. 705–716.
- [11] Alicia DeVos, Aditi Dhabalia, Hong Shen, Kenneth Holstein, and Motahhare Eslami. 2022. Toward User-Driven Algorithm Auditing: Investigating users' strategies for uncovering harmful algorithmic behavior. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [12] Alicia DeVos, Aditi Dhabalia, Hong Shen, Kenneth Holstein, and Motahhare Eslami. 2022. Toward User-Driven Algorithm Auditing: Investigating users' strategies for uncovering harmful algorithmic behavior. *CHI Conference on Human Factors in Computing Systems* (2022).
- [13] Yao Dou, Isadora Krsek, Tarek Naous, Anubha Kabra, Sauvik Das, Alan Ritter, and Wei Xu. 2024. Reducing Privacy Risks in Online Self-Disclosures with Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 13732–13754. <https://doi.org/10.18653/v1/2024.acl-long.741>
- [14] Motahhare Eslami, Aimee Rickman, Kristen Vaccaro, Amirhossein Aleyasen, Andy Vuong, Karrie Karahalios, Kevin Hamilton, and Christian Sandvig. 2015. "I always assumed that I wasn't really that close to [her]" Reasoning about Invisible Algorithms in News Feeds. In *Proceedings of the 33rd annual ACM conference on human factors in computing systems*. 153–162.
- [15] Eve Fleisig, Aubrie Amstutz, Chad Atalla, Su Lin Blodgett, Hal Daumé III, Alexandra Olteanu, Emily Sheng, Dan Vann, and Hanna Wallach. 2023. Fair-Prism: Evaluating fairness-related harms in text generation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics.
- [16] Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, et al. 2022. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858* (2022).
- [17] Timnit Gebru. 2021. Hierarchy of Knowledge in Machine Learning and Related Fields and Its Consequences. <https://www.youtube.com/watch?v=OL3DowBM9uc>
- [18] Sebastian Gehrmann, Elizabeth Clark, and Thibault Sellam. 2023. Repairing the cracked foundation: A survey of obstacles in evaluation practices for generated text. *Journal of Artificial Intelligence Research* 77 (2023), 103–166.
- [19] Ahmad Ghazal, Tilmann Rabl, Mingqing Hu, Francois Raab, Meikel Poess, Alain Crolotte, and Hans-Arno Jacobsen. 2013. Bigbench: Towards an industry standard benchmark for big data analytics. In *Proceedings of the 2013 ACM SIGMOD international conference on Management of data*. 1197–1208.
- [20] David Heineman, Yao Dou, Mounica Maddela, and Wei Xu. 2023. Dancing Between Success and Failure: Edit-level Simplification Evaluation using SALSA. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 3466–3495. <https://doi.org/10.18653/v1/2023.emnlp-main.211>
- [21] David Heineman, Yao Dou, and Wei Xu. 2023. Thresh: A Unified, Customizable and Deployable Platform for Fine-Grained Text Evaluation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Yansong Feng and Els Lefever (Eds.). Association for Computational Linguistics, Singapore, 336–345. <https://doi.org/10.18653/v1/2023.emnlp-demo.30>
- [22] Kenneth Holstein, Jennifer Wortman Vaughan, Hal Daumé III, Miro Dudik, and Hanna Wallach. 2019. Improving fairness in machine learning systems: What do industry practitioners need?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [23] Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Pratik Ringshia, et al. 2021. Dynabench: Rethinking benchmarking in NLP. *arXiv preprint arXiv:2104.14337* (2021).
- [24] Tae Soo Kim, Yoonjoo Lee, Minsuk Chang, and Juho Kim. 2023. Cells, Generators, and Lenses: Design Framework for Object-Oriented Interaction with Large Language Models. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST'23)*.
- [25] Tae Soo Kim, Yoonjoo Lee, Jamin Shin, Young-Ho Kim, and Juho Kim. 2023. EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined Criteria. *arXiv preprint arXiv:2309.13633* (2023).
- [26] Michelle S. Lam, Mitchell L. Gordon, Danaë Metaxa, Jeffrey T. Hancock, James A. Landay, and Michael S. Bernstein. 2022. End-User Audits: A System Empowering Communities to Lead Large-Scale Investigations of Harmful Algorithmic Behavior. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 512 (Nov 2022), 34 pages. <https://doi.org/10.1145/3555625>
- [27] Michelle S. Lam, Ayush Pandit, Colin H. Kalicki, Rachit Gupta, Poonam Sahoo, and Dana'e Metaxa. 2023. Sociotechnical Audits: Broadening the Algorithm Auditing Lens to Investigate Targeted Advertising. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 360 (Oct 2023), 37 pages. <https://doi.org/10.1145/3610209>
- [28] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. 2022. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110* (2022).
- [29] Q Vera Liao, Daniel Gruen, and Sarah Miller. 2020. Questioning the AI: informing design practices for explainable AI user experiences. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–15.
- [30] Q Vera Liao and S Shyam Sundar. 2022. Designing for responsible trust in AI systems: A communication perspective. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*. 1257–1268.
- [31] Q Vera Liao and Jennifer Wortman Vaughan. 2023. AI Transparency in the Age of LLMs: A Human-Centered Research Roadmap. *arXiv preprint arXiv:2306.01941* (2023).
- [32] Q Vera Liao and Ziang Xiao. 2023. Rethinking Model Evaluation as Narrowing the Socio-Technical Gap. *arXiv preprint arXiv:2306.03100* (2023).
- [33] Daniel Liebling, Katherine Heller, Samantha Robertson, and Wesley Deng. 2022. Opportunities for Human-centered Evaluation of Machine Translation Systems. In *Findings of the Association for Computational Linguistics: NAACL 2022*. 229–240.
- [34] Yu Lu Liu, Su Lin Blodgett, Jackie Cheung, Q. Vera Liao, Alexandra Olteanu, and Ziang Xiao. 2024. ECBD: Evidence-Centered Benchmark Design for NLP. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 16349–16365. <https://doi.org/10.18653/v1/2024.acl-long.861>
- [35] Yu Lu Liu, Meng Cao, Su Lin Blodgett, Jackie Chi Kit Cheung, Alexandra Olteanu, and Adam Trischler. 2023. Responsible AI Considerations in Text Summarization Research: A Review of Current Practices. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 6246–6261. <https://doi.org/10.18653/v1/2023.findings-emnlp.413>
- [36] Michael Madaio, Lisa Egede, Hariharan Subramonyam, Jennifer Wortman Vaughan, and Hanna Wallach. 2021. Assessing the Fairness of AI Systems: AI Practitioners' Processes, Challenges, and Needs for Support. *arXiv preprint arXiv:2112.05675* (2021).
- [37] Mounica Maddela, Yao Dou, David Heineman, and Wei Xu. 2023. LENS: A Learnable Evaluation Metric for Text Simplification. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long*

- Papers), Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (Eds.). Association for Computational Linguistics, Toronto, Canada, 16383–16408. <https://doi.org/10.18653/v1/2023.acl-long.905>
- [38] Ethan Mendes, Yang Chen, Wei Xu, and Alan Ritter. 2023. Human-in-the-loop Evaluation for Early Misinformation Detection: A Case Study of COVID-19 Treatments. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (Eds.). Association for Computational Linguistics, Toronto, Canada, 15817–15835. <https://doi.org/10.18653/v1/2023.acl-long.881>
- [39] Danaë Metaxa, Joon Sung Park, Ronald E Robertson, Karrie Karahalios, Christo Wilson, Jeff Hancock, Christian Sandvig, et al. 2021. Auditing algorithms: Understanding algorithmic systems from the outside in. *Foundations and Trends® in Human-Computer Interaction* 14, 4 (2021), 272–344.
- [40] Tarek Naous, Michael Ryan, Alan Ritter, and Wei Xu. 2024. Having Beer after Prayer? Measuring Cultural Bias in Large Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 16366–16393. <https://doi.org/10.18653/v1/2024.acl-long.862>
- [41] Besmira Nushi, Ece Kamar, and Eric Horvitz. 2018. Towards accountable ai: Hybrid human-machine analyses for characterizing system failure. In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, Vol. 6. 126–135.
- [42] Rodrigo Ochigame and Katherine Ye. 2021. Search Atlas: Visualizing Divergent Search Results Across Geopolitical Borders. In *Designing Interactive Systems Conference 2021. 1970–1983*.
- [43] Inioluwa Deborah Raji, Andrew Smart, Rebecca N White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. 2020. Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*. 33–44.
- [44] Inioluwa Deborah Raji, Peggy Xu, Colleen Honigsberg, and Daniel Ho. 2022. Outsider Oversight: Designing a Third Party Audit Ecosystem for AI Governance. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*. 557–571.
- [45] Charvi Rastogi, Marco Tulio Ribeiro, Nicholas King, and Saleema Amershi. 2023. Supporting Human-AI Collaboration in Auditing LLMs with LLMs. *arXiv preprint arXiv:2304.09991* (2023).
- [46] Christian Sandvig, Kevin Hamilton, Karrie Karahalios, and Cedric Langbort. 2014. Auditing algorithms: Research methods for detecting discrimination on internet platforms. *Data and Discrimination: Converting Critical Concerns into Productive Inquiry* (2014).
- [47] Andrew D Selbst, Danah Boyd, Sorelle A Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and abstraction in sociotechnical systems. In *Proceedings of the conference on fairness, accountability, and transparency*. 59–68.
- [48] Hong Shen, Wesley. H. Deng, Aditi Chattopadhyay, Zhiwei Steven Wu, Xu Wang, and Haiyi Zhu. 2021. Value cards: An educational toolkit for teaching social impacts of machine learning through deliberation. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*. 850–861.
- [49] Hong Shen, Leijie Wang, Wesley. H. Deng, Ciell Brusse, Ronald Velgersdijk, and Haiyi Zhu. 2022. The Model Card Authoring Toolkit: Toward Community-centered, Deliberation-driven AI Design. In *2022 ACM Conference on Fairness, Accountability, and Transparency*. 440–451.
- [50] Mohammad Tahaei, Marios Constantinides, Daniele Quercia, Sean Kennedy, Michael Muller, Simone Stumpf, Q Vera Liao, Ricardo Baeza-Yates, Lora Aroyo, Jess Holbrook, et al. 2023. Human-Centered Responsible Artificial Intelligence: Current & Future Trends. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–4.
- [51] Laura Weidinger, Jonathan Uesato, Maribeth Rauh, Conor Griffin, Po-Sen Huang, John Mellor, Amelia Glaese, Myra Cheng, Borja Balle, Atoosa Kasirzadeh, et al. 2022. Taxonomy of risks posed by language models. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*. 214–229.
- [52] Ziang Xiao, Wesley Hanwen Deng, Michelle S Lam, Motahhare Eslami, Juho Kim, Mina Lee, and Q Vera Liao. 2024. Human-Centered Evaluation and Auditing of Language Models. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–6.
- [53] Ziang Xiao, Susu Zhang, Vivian Lai, and Q Vera Liao. 2023. Evaluating NLG Evaluation Metrics: A Measurement Theory Perspective. *arXiv preprint arXiv:2305.14889* (2023).
- [54] Ziang Xiao, Michelle X Zhou, Q Vera Liao, Gloria Mark, Changyan Chi, Wenxi Chen, and Huahai Yang. 2020. Tell me about yourself: Using an AI-powered chatbot to conduct conversational surveys with open-ended questions. *ACM Transactions on Computer-Human Interaction (TOCHI)* 27, 3 (2020), 1–37.
- [55] Seonghyeon Ye, Doyoung Kim, Sungdong Kim, Hyeonbin Hwang, Seungone Kim, Yongrae Jo, James Thorne, Juho Kim, and Minjoon Seo. 2023. Flask: Fine-grained language model evaluation based on alignment skill sets. *arXiv preprint arXiv:2307.10928* (2023).
- [56] Meg Young, Lassana Magassa, and Batya Friedman. 2019. Toward inclusive tech policy design: a method for underrepresented voices to strengthen tech policy documents. *Ethics and Information Technology* 21, 2 (2019), 89–103.
- [57] Mireia Yurrita, Tim Draws, Agathe Balayn, Dave Murray-Rust, Nava Tintarev, and Alessandro Bozzon. 2023. Disentangling Fairness Perceptions in Algorithmic Decision-Making: the Effects of Explanations, Human Oversight, and Contestability. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–21.
- [58] Yunfeng Zhang, Q Vera Liao, and Rachel KE Bellamy. 2020. Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*. 295–305.
- [59] Shi Zong, Ashutosh Baheti, Wei Xu, and Alan Ritter. 2022. Extracting a Knowledge Base of COVID-19 Events from Social Media. In *Proceedings of the 29th International Conference on Computational Linguistics*. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, 3810–3823. <https://aclanthology.org/2022.coling-1.335>